capacity to keep track of disorder *exactly* equals the maximum disorder a region can possibly contain, not more and not less.

The thing is, if the Tupperware conclusion had had unlimited validity, a fundamental theory would have needed the capacity to account for a volume's worth of disorder in any region. But since that reasoning fails when gravity is included—and since a fundamental theory must include gravity—we learn that a fundamental theory need only be able to account for a surface area's worth of disorder in any region. And as we showed with a couple of numerical examples a few paragraphs ago, for large regions the latter is much smaller than the former.

Thus, the Bekenstein and Hawking result tells us that a theory that includes gravity is, in some sense, simpler than a theory that doesn't. There are fewer "degrees of freedom"—fewer things that can change and hence contribute to disorder—that the theory must describe. This is an interesting realization in its own right, but if we follow this line of reasoning one step further, it seems to tell us something exceedingly bizarre. If the maximum entropy in any given region of space is proportional to the region's surface area and not its volume, then perhaps the true, fundamental degrees of freedom—the attributes that have the potential to give rise to that disorder—*actually reside on the region's surface and not within its volume.* Maybe, that is, the universe's real physical processes take place on a thin, distant surface that surrounds us, and all we see and experience is merely a projection of those processes. Maybe, that is, the universe is rather like a hologram.

This is an odd idea, but as we'll now discuss, it has recently received substantial support.

Is the Universe a Hologram?

A hologram is a two-dimensional piece of etched plastic, which, when illuminated with appropriate laser light, projects a three-dimensional image.[6] In the early 1990s, the Dutch Nobel laureate Gerard 't Hooft and Leonard Susskind, the same physicist who coinvented string theory, suggested that the universe itself might operate in a manner analogous to a hologram. They put forward the startling idea that the comings and